

# SHI FENG

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## EDUCATION

### Tsinghua University

Aug 2019 - Jul 2023 (expected)

#### Bachelor of Science, selected to Yao Class

- GPA: 3.87/4.0
- GRE Scores: Verbal Reasoning 157, Quantitative Reasoning 170, Analytical Writing 4.0
- TOEFL Best Scores: Reading 30, Listening 29, Speaking 22, Writing 29
- Selected Courses: Introduction to AI (A), Mathematics for Computer Science (A), Causal and Statistical Learning (A+), Abstract Algebra (A), Introduction to Databases (A+), Artificial Intelligence: Principles and Techniques (A), Theory of Computation (A), Distributed and Blockchain System (A-), Machine Learning (A), Operating System (A), Research Immersion Training (A+)

## RESEARCH INTERESTS

Theoretical computer science and economics, particularly algorithmic game theory, learning theory, causality, and machine learning.

## RESEARCH EXPERIENCES

### Multi-Agent Owner-Assisted Scoring Mechanisms

Aug 2022 - present

Advisors: Prof. [Yuhao Wang](#), Prof. [Weijie Su](#)

Tsinghua IIIS&UPenn Wharton School

- Extended owner-assisted scoring problem to a multi-agent setting and wrote a research proposal for it.
- Designed a fair, “exchangeable noise assumption free”, and truthful mechanism for the extended problem.

### Peer Prediction for Learning Agents

Feb 2022 - Jun 2022

Advisors: Prof. [Yiling Chen](#), Prof. [Fang-Yi Yu](#)

Harvard EconCS Group (in-person)

- Surveyed mechanisms for information elicitation and extended the original multi-task setting to a sequential setting.
- Proved truthful convergence of correlated agreement mechanism adopted on agents using reward-based online learning algorithms and a negative result showing no-regret assumption on agents can-not guarantee truthful convergence.
- Created a peer prediction simulator and conducted numerical experiments which supported theoretical results.

### Combinatorial Causal Bandits

Jul 2021 - Feb 2022

Advisor: Prof. [Wei Chen](#)

MSR Asia Theory Center

- Surveyed causal bandits algorithms for pure exploration and cumulative regret minimization.
- Presented the new combinatorial causal bandit framework with a definition of cumulative regret.
- Proposed algorithms for combinatorial causal bandits problem on binary generalized linear models with hidden variables.
- Implemented algorithms in OOP and conducted numerical experiments to show their efficiency.

### Causal Inference for Social Networks

Nov 2020 - Jul 2021

Advisor: Prof. Wei Chen

MSR Asia Theory Center

- Surveyed identifiability of Bayesian causal models and completeness of the do-calculus algorithm.
- Provided transformations from an Independent Cascade Model to a Bayesian causal model.
- Proved necessary and sufficient conditions for identifiability in three common types of Independent Cascade Models.

## PUBLICATIONS

(\* denotes equal contribution)

**Shi Feng**, Wei Chen. Combinatorial Causal Bandits.

The 37th AAAI Conference on Artificial Intelligence (AAAI 2023).

[\[arXiv\]](#) [\[code\]](#)

**Shi Feng**, Fang-Yi Yu, Yiling Chen. Peer Prediction for Learning Agents.

The 36th Conference on Neural Information Processing Systems (NeurIPS 2022).

[\[arXiv\]](#) [\[slides\]](#) [\[code\]](#)

**Shi Feng**, Wei Chen. Causal Inference for Influence Propagation–Identifiability of the Independent Cascade Model.

The 10th International Conference on Computational Data and Social Networks (CSoNet 2021) **Best Paper**.

[\[arXiv\]](#) [\[slides\]](#) [\[publication\]](#)

**Shi Feng\***, Zimeng Song\*, Weijie Su, Yuhao Wang. A Truthful Multi-Agent Owner-Assisted Scoring Mechanism.

Writing in progress.

[\[slides\]](#)

## SELECTED HONORS & AWARDS

Comprehensive Excellence Awards (top 10%)

Oct 2020, Oct 2022, THU

“Star of Tomorrow” Award of Excellence

Mar 2022, MSR Asia

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|-------------------------------------------------------------------------------------|---------------------------------------------|
| <b>Best Paper Award</b> (1/57)                                                      | <i>Nov 2021, CSoNet</i>                     |
| <b>Innovation Excellence Award</b>                                                  | <i>Oct 2021, THU</i>                        |
| <b>Social Practice Excellence Award</b>                                             | <i>Oct 2021, THU</i>                        |
| <b>“Challenge Cup” Science and Technology Competition, Top Grade Prize</b> (top 3%) | <i>Apr 2021, THU</i>                        |
| <b>Freshman Scholarship</b> (top 10%)                                               | <i>Oct 2019, THU</i>                        |
| <b>Tsinghua Xuetaang Scholarship</b>                                                | <i>Aug 2019, THU</i>                        |
| <b>Russian Mathematical Olympiad, Silver Medal</b>                                  | <i>Apr 2019, MSE of Russia</i>              |
| <b>Chinese Mathematical Olympiad, Gold Medal</b>                                    | <i>Nov 2018, CMS</i>                        |
| <b>National High School Mathematics League, First Prizes</b>                        | <i>Sept 2016, Sept 2017, Sept 2018, CMS</i> |

## TALKS

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|----------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>NeurIPS 2022</b> , <a href="#">Peer Prediction for Learning Agents</a>                                                        | <i>Oct 2022</i>  |
| <b>Yao Class Seminar</b> , <a href="#">Peer Prediction for Learning Agents</a>                                                   | <i>Sept 2022</i> |
| <b>CSoNet 2021</b> , <a href="#">Causal Inference for Influence Propagation–Identifiability of the Independent Cascade Model</a> | <i>Nov 2021</i>  |

## SERVICES

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|------------------------------------------------------------------------------|---------------------------|
| <b>Class Leader</b> , Yao Class 91, Tsinghua University                      | <i>Aug 2022 - present</i> |
| <b>Reviewer</b> , Journal of Combinatorial Optimization (JOCO)               | <i>May 2022</i>           |
| <b>Team Member</b> , Chinese national team for Russian Mathematical Olympiad | <i>Apr 2019</i>           |

## SKILLS

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**Programming Languages:** C/C++, Python, Golang, SQL, assembly language, Bash, HTML

**Professional Applications:** Latex, Mathematica, Matlab, PyTorch, TensorFlow, Git

**Languages:** Chinese (native language), English (fluent)

**Hobbies:** soccer (member of IIIS soccer team), road trip (driving through China National Highway 318 in 2019), handball (amateur training for one year), basketball, vehicle engineering (1 invention patent and 1 utility model patent), Go (amateur 1 dan)